

Ho Han Sheng

 ho-han-sheng |  GitHub |  hsho |  hsho005@suss.edu.sg |  0009-0000-8091-4045

WORK EXPERIENCE

Research Assistant

Oct 2023 - Present

School of Humanities and Behavioural Sciences, Singapore University of Social Sciences

Primary: [Research in Psychophysiology, Quality of Life, Stress, & Time Use \(PQRST\) Lab](#)

- PI: A/Prof Emily Ortega
- Lab manager and supervision of junior RAs
- Conducted data cleaning, pre-processing, and statistical analyses in R for the National Council of Social Service - Ngee Ann Kongsi (NCSS-NAK) 360 Panel Study
- Conceptualised and applied mixed graphical models for exploratory data analysis of high-dimensional datasets
- Typeset large internal reports (≈ 400 pages) with $\text{\LaTeX}$, including tables, figures, and appendices
- Unpublished results informing policy-making on: (a) gender disparities in unpaid work, (b) work-life balance and flexible work arrangements, and (c) quality of life past COVID-19
- Co-written grant applications

Secondary: [Affect, Ambivalence, & Adjustment \(AAA\) Lab](#)

- PI: Dr. Oh Yong Sheng, Vincent
- Lab manager, supervision of junior RAs and honours capstone students
- Lead RA and collaborator for ADAPTIVE study
- Supervised a group research project comprising four undergraduate volunteers conducting secondary data analyses of SHARE-ERIC and CHARLS
- Engineered a physical slider response box using Arduino for experimental use in affective studies
- Co-wrote grant and IRB applications
- Invited teaching assistant for PSY405: Applied Research in Social Psychology

Tertiary: [The Relational Brain Lab](#)

- PI: A/Prof Atiqah Azhari
- Implemented pre-processing and analytical pipeline for fNIRS data in Python
- Conceptualisation and formal analysis of neural synchrony through vector-autoregressive methods

Quaternary: Bilingual Language and Thought Lab

- PI: Dr. Ooi Seok Hui
- Drafting IRB applications and preregistrations
- Provide formal analysis support on Structural Equation Modelling (SEM)

Research Assistant

Jul 2025 - Present

Research Cluster, School of Humanities and Behavioural Sciences, Singapore University of Social Sciences

- PI: A/Prof Atiqah Azhari
- Served as SHBS Behavioural Sciences Ethics Review Committee secretariat
- Provided research support services (e.g., research design, formal analysis, report writing) to wider School of Humanities and Behavioural Sciences faculty
- Provided ad-hoc administrative support

Research Intern

Jan 2023 - Jul 2023

Multimodal Neuroimaging in Neuropsychiatric Disorders Laboratory, Center for Sleep and Cognition, Yong Loo Lin School of Medicine, National University of Singapore

- Utilised Python and R for mixture modelling of eye-tracking data, data cleaning, and exploratory data analysis
- Conducted qualitative interviews for stimulus recall tasks and scored neurocognitive batteries

Peer Mentor

Jan 2022 - Dec 2022

School of Science and Technology, Singapore University of Social Sciences

- Conducted weekly tutorials for MTH219: Fundamentals of Statistics and Probability and MTH220: Statistical Methods and Inference

GRANTS

Affective Dynamics, Attrition, and Psychophysiology: Targeted Interventions Via Experience-sampling (ADAPTIVE)

2026 - 2029

- Collaborator (PI: Dr Oh Yong Sheng, Vincent)
- Ministry of Education Academic Research Fund Tier 1: \$72,762

CONFERENCE PRESENTATIONS

Ho, H. S., Ortega, E., Chan, Y. X. B., Tay, X. A., & Tang, S. O. S. (2024, September 2–6). *Exploratory network modelling of complex human systems in time-use research: Results from wave 1 of the Singapore NCSS-NAK 360 Panel Study* [Conference presentation]. Conference on Complex Systems 2024, Exeter, England.

Tay, X. A., Ortega, E., Chan, Y. X. B., **Ho, H. S.**, & Tang, S. O. S. (2024, November 6–7). *Families who eat together stay together?: Examining time use of families in Singapore by family structure* [Conference presentation]. Asian Family Conference 2024, Singapore.

Ho, H. S., Ortega, E., & Oh, V. Y. S. (2025, December 1–3). *Psychophysiological and temporal dynamics of mixed emotions* [Conference presentation]. 33rd Annual Meeting of the Australasian Society for Psychophysiology (ASP2025), Canberra, Australia.

Tan, P. L., Gee, C., Ortega, E., Tay, X. A., **Ho, H. S.**, & Tan, D. (2026, May 21–22). *Singapore National Time Transfer Accounts: Life cycle of non-market economic production* [Conference paper]. PAS 2026 Annual Conference, Singapore.

PUBLICATIONS

Ortega, E., Tay, X. A., **Ho, H. S.**, Chan, Y. X. B., Cheok, S. Z. M., & Tang, S. O. S. (2024, November 7). *NCSS-NAK 360 Panel Research: Wave 1 report* [Unpublished internal report]. National Council of Social Service.

Ortega, E., Tay, X. A., **Ho, H. S.**, Tang, S. O. S., & Chan, Y. X. B. (2025a, March 21). *NCSS-NAK 360 Panel Study: Wave 2 report* [Unpublished internal report]. National Council of Social Service.

Ortega, E., Tay, X. A., **Ho, H. S.**, Tang, S. O. S., & Chan, Y. X. B. (2025b, May 30). *NCSS-NAK 360 Panel Study: Wave 3 report* [Unpublished internal report]. National Council of Social Service.

- Azhari, A., Rai, A., **Ho, H. S.**, Jegathisan, A., Gill, S., Norfor, N., & Chen, Z. (2025). Inter-brain synchrony between undergraduate students during a naturalistic online seminar predicted greater relational satisfaction and task performance. *Frontiers in Neuroscience*, 19, Article 1705767. <https://doi.org/10.3389/fnins.2025.1705767>
- Ortega, E., Tay, X. A., Shu, H., Lee, W. P. R., Chan, Y. X. B., **Ho, H. S.**, & Tang, S. O. S. (2025). *Gender differences in how we spend our time: A Singapore perspective* [Manuscript in preparation]. School of Humanities and Behavioural Sciences, Singapore University of Social Sciences.
- Ho, H. S.** & Ortega, E. (2025). *A network psychometrics approach to analysing heart rate variability* [Manuscript in preparation]. School of Humanities and Behavioural Sciences, Singapore University of Social Sciences.

PROJECTS

Affective Dynamics, Attrition, and Psychophysiology: Targeted Interventions Via Experience-sampling (ADAPTIVE) **Ongoing**

- CRediT contributor roles: Conceptualisation, funding acquisition, methodology, project administration, supervision
- Developed grant proposals, IRB applications, budgets and fund allocation
- Liaison with external research collaborators
- Conceptualised intensive longitudinal, experience sampling protocol integrated with continuous wearable measurement ([TechDoctor](#))
- Implemented planned-missingness, anchor-item design for reducing respondent burden
- Project aims to investigate: (a) how various psychological and physiological factors predict academic and well-being outcomes; (b) whether a targeted intervention ([intellect.co](#)) can be implemented to help mitigate the academic strains that students face, improve academic and well-being outcomes; and (c) reduce intentions to drop out from university

Biopsychosocial Determinants of Prejudice **Ongoing**

- Named collaborator (PI: Dr Oh Yong Sheng, Vincent)
- Pilot grant proposal submitted in November 2025
- Developed experimental protocol integrating psychophysiological measures, continuous self-reporting, and neuroimaging
- Responsible for conducting formal analysis of intensive longitudinal data with network psychometrics, vector-autoregressive models, and GIMME
- This project integrates biopsychosocial components to examine how prejudice develops toward marginalised groups (e.g., the elderly, immigrants, racial minorities), and evaluates the efficacy of targeted interventions in buffering against prejudicial content exposure

Linguistic distance and bilingual executive control **Ongoing**

- PI: Dr. Ooi Seok Hui
- Drafting pre-registration of secondary data analysis
- Project aims to: (a) conceptualise linguistic distance as a latent factor and (b) examine its effects on executive control

Psychophysiological and Temporal Dynamics of Mixed Emotions

Ongoing

- Honours capstone project (Co-Supervisors: A/Prof Emily Ortega; Dr. Oh Yong Sheng, Vincent)
- Integrated psychophysiological measures (EmotiBit), continuous self-reported valence (via game controller), cognitive appraisals, and video stimuli in PsychoPy experiment
- Developing idiographic, dynamical models of emotional experiences with TV-VAR
- Comparing (in)variance in emotional conditions with INIT and group-level similarity via mlVAR and GIMME

Effectiveness of Heart Rate Variability Biofeedback Training on Patients Undergoing Cardiac Rehabilitation (HRV-CARE)

Ongoing

- Conceptualised longitudinal panel research design for randomised control trial
- Devised Structural Equation Modelling (SEM) and Generalised Network Psychometrics analysis plan to examine multimodal psychophysiological and behavioural measures
- Integrated wearable data collection platforms ([TechDoctor](#)) to capture continuous change in biomarkers across study duration
- Supervised team of junior part-time research assistants

Experiences with Quantitative Methods in the Undergraduate Psychology Curriculum

Ongoing

- Extension of independent research submitted for PSY455: Qualitative Methods
- Phenomenological study on psychology undergraduates' experiences learning quantitative methods
- Findings informed ongoing redesign of undergraduate quantitative methods curriculum
- Implemented [OpenAI's Whisper](#) for transcription and QualCoder for analysis

Timing of Help and Future Motivation: The Role of Self-Efficacy in Singapore's Academic Context

Jan 2025 - Apr 2025

- Group research project under PSY405: Applied Research in Social Psychology
- Devised mathematical specifications for hypotheses
- Conceptualised project research design and experimental procedures
- Conducted formal analysis (through R) with ANCOVA in a Latent Change Score specification for modelling mediation in a two-instance, pretest-posttest experimental design
- Wrote and interpreted statistical results through an SEM framework

Mental Wellness Toolkit @ North West CDC

2022 – 2023

- Conducted literature review on stress
- Authored sections on stress and coping strategies

PsyConnect Statistics Support Group

May 2022 – present

- Founded and coordinated a peer-run statistics consulting platform for SUSS Psychology students

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6 Dec 2025

- Development and teaching of an introductory R programming workshop for SUSS Psychology students

AWARDS & HONOURS

Exemplary SUSS Psychology Student Award, Singapore Psychological Society **2026**

- Conferred to one psychology undergraduate from each IHL, who has demonstrated exemplary character and outstanding personal qualities through contribution to the society at-large, in addition to academic and research excellence in the field of psychology.

Student Travel Award, Singapore University of Social Sciences **2025**

- $\approx$ \$3100 SGD
- For oral presentation “Psychophysiological and Temporal Dynamics of Mixed Emotions” at 33rd Annual Meeting of the Australasian Society for Psychophysiology (ASP2025)

Travel Award, Australasian Society for Psychophysiology **2025**

- $\approx$ \$350 AUD
- For oral presentation “Psychophysiological and Temporal Dynamics of Mixed Emotions” at 33rd Annual Meeting of the Australasian Society for Psychophysiology (ASP2025)

Student Travel Award, Singapore University of Social Sciences **2024**

- $\approx$ \$4700 SGD
- For contributed talk “Exploratory Network Modelling of Complex Human Systems in Time-Use Research: Results from Wave 1 of the Singapore NCSS-NAK 360 Panel Study” at Conference on Complex Systems 2024

Dean’s List, Singapore University of Social Sciences **2024**

- For excellent performance in the BSc Psychology programme
- Academic accomplishments indicating attainment of a high level of competence and self-discipline

SUSS Community Impact Award, Singapore University of Social Sciences **2023**

- For “Mental Wellness Toolkit @ North West CDC”
- Recognises and spotlights the positive impact of SUSS students’ collaborative efforts within the community

SKILLS

Tools $\LaTeX$, Git, PsychoPy, [EmotiBit](#), Arduino, [QualCoder](#), SPSS, Qualtrics
Programming Languages R, Python

EDUCATION

Bachelor of Science (Honours), Psychology **2025**

Singapore University of Social Sciences

Professional Certificate in Network Psychometrics for Behavioural and Social Scientists **Aug 2024**

National University of Singapore